

# Physical Chemistry of Materials

## Lecture 09

1차 중간고사 안내 및 복습



# 1차 중간고사 안내

- 시간: 10월 15일(화) 오전 10:30~11:30
- 장소: 화학관 112호(홀수 학번), 414호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

# 시험 범위

기초 열역학(2강)

상 전이(3강)

통계역학의 원리(4강)

정준 앙상블의 응용(5강)

이상 기체/화학 평형(6강)

상호작용하는 입자들(7강)

대정준 앙상블(8강)

# 시험 문제

총 여섯 문제

개념을 묻는 문제: 두 문제

공식을 유도하는 문제: 세 문제

응용 문제: 한 문제

# 소재물리화학의 구성

## 고전 열역학

열역학 복습

상 전이

## 통계역학

기초 이론

이상 기체

상호작용 계

## 고체

고체의 구조

고체의 진동

고체의 전기적 성질

고체의 자기적 성질

표면화학

나노화학

## 연성물질

액체

유리

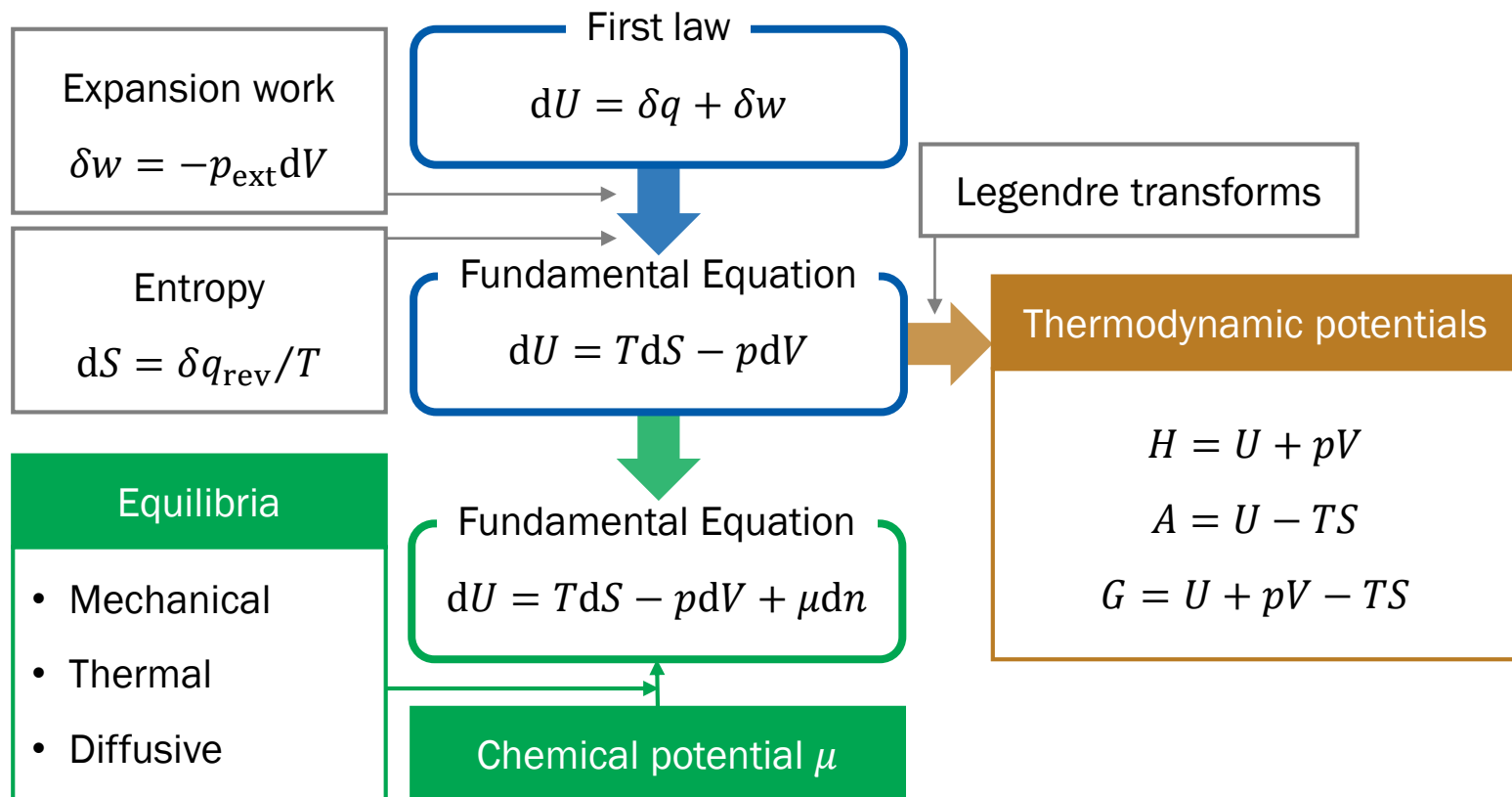
고분자의 구조

고분자의 상 거동

고분자의 젤화

액정, 콜로이드, ...

# Basic Thermodynamics



# Phase Transitions

- Stability of phases: stable, metastable, unstable
- Binodal and spinodal
- Example: phase separation
  - Phase diagram
  - Kinetics: spinodal decomposition vs. nucleation and growth
- Nucleation and growth models
  - Classical nucleation theory
  - Ostwald ripening

# Statistical Mechanics

Ensemble



Principle of maximum entropy

$$S = -k_B \sum_i p_i \ln p_i$$



Constrained optimization



Determination of multipliers  $(\lambda, \beta, \alpha, \dots)$



# Ensembles

|                         | Microcanonical                          | Canonical                                         | Grand canonical                                                                |
|-------------------------|-----------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|
| Fluctuation             | No                                      | $E$                                               | $E$ and $N$                                                                    |
| $p_i$                   | $1/W$                                   | $e^{-\beta E_i}/Z$                                | $e^{-\beta E_i - \alpha N_i}/\mathcal{Z}$                                      |
| Normalization constant  | Number of microstates<br>$W = \sum_i 1$ | Partition function<br>$Z = \sum_i e^{-\beta E_i}$ | Grand partition function<br>$\mathcal{Z} = \sum_i e^{-\beta E_i - \alpha N_i}$ |
| Multipliers             |                                         | $\beta = 1/k_B T$                                 | $\beta = 1/k_B T$<br>$\alpha = -\tilde{\mu}/k_B T$                             |
| Thermodynamic potential | Entropy<br>$S = k_B \ln W$              | Helmholtz energy<br>$A = -k_B T \ln Z$            | Grand potential<br>$\phi = -k_B T \ln \mathcal{Z}$                             |

# Useful Trick

For  $p_i = C_i e^{-kx_i} / Z$  and  $Z = \sum_i C_i e^{-kx_i}$ ,

$$\langle x \rangle = \sum_i x_i p_i = \frac{1}{Z} \sum_i x_i C_i e^{-kx_i}$$

but

$$\frac{\partial \ln Z}{\partial k} = \frac{1}{Z} \frac{\partial Z}{\partial k} = \frac{1}{Z} \frac{\partial}{\partial k} \sum_i C_i e^{-kx_i} = -\frac{1}{Z} \sum_i x_i C_i e^{-kx_i}$$

Hence,

$$\langle x \rangle = -\frac{\partial \ln Z}{\partial k}$$

# 1-Particle Partition Functions

- Discrete microstates

$$z = \sum_s e^{-\beta E(s)}$$

- Classical particles (in 1D)

$$z = \frac{1}{h} \iint e^{-\beta E(x,p)} dx dp$$

# $N$ -Particle Partition Functions

- Independent & distinguishable particles

$$Z = z^N \text{ (} z: \text{ one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: depends on interactions

# Applications: Canonical Ensemble

- Ideal gas: QM vs. CM
  - Translation – particle in a box
  - Vibration – simple harmonic oscillator
  - Rotation – rigid rotor
- Equilibrium constant of ideal gas
- Real gas
  - Intramolecular interactions
  - Evaluation of partition function



# Grand Partition Functions

$$\mathcal{Z} = \sum_i e^{-\alpha N_i - \beta E_i}$$

$$\mathcal{Z} = \sum_N e^{-\alpha N} \left( \sum_s e^{-\beta E_s(N)} \right)$$

For independent particles,  $E_s(N) = N\epsilon_s$

$$\mathcal{Z} = \sum_s \sum_N e^{-\alpha N - \beta N \epsilon_s}$$

# Calculation of $\mathcal{Z}$

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N Z(N)$$

- Independent & distinguishable particles

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N z^N = \sum_N (ze^{\mu/RT})^N$$

- Independent & indistinguishable particles

$$\mathcal{Z} = \sum_N (e^{\mu/RT})^N \frac{z^N}{N!} = \sum_N \frac{1}{N!} (ze^{\mu/RT})^N = \exp(ze^{\mu/RT})$$

# Specific 1-Particle Microstate

$$\mathcal{Z} = \sum_s \sum_N e^{-\alpha N - \beta N \epsilon_s} = \sum_s \mathcal{Z}_s$$

For specific 1-particle microstate  $s$ ,

$$\mathcal{Z}_s = \sum_N e^{-\alpha N - \beta N \epsilon_s}$$

- **Fermion:**  $N = 0$  or  $1$
- **Boson:**  $N =$  any nonnegative integer  $(0, 1, 2, \dots)$



# Average Occupation Number

For a specific microstate with energy  $\epsilon$ ,

- Classical particles (Maxwell-Boltzmann distribution)

$$N_{\text{MB}}(\epsilon) = e^{-(\epsilon - \tilde{\mu})/k_{\text{B}}T}$$

- Fermions (Fermi-Dirac distribution)

$$N_{\text{FD}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} + 1}$$

- Bosons (Bose-Einstein distribution)

$$N_{\text{BE}}(\epsilon) = \frac{1}{e^{(\epsilon - \tilde{\mu})/k_{\text{B}}T} - 1}$$